FRAP Analysis Postprocessing Pipeline

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1 Overview

The FRAP post-analysis pipeline transforms raw target and control fluorescence measurements into corrected recovery curves, summarizes them at the group level, fits a one-phase exponential recovery model, and derives quantitative FRAP parameters such as recovery half-time, mobile fraction, and immobile fraction.

The complete processing chain can be summarized as

$$\text{Raw intensities} \rightarrow \text{Photobleaching correction} \rightarrow \text{Normalization} \rightarrow$$

$$\rightarrow \text{Group mean recovery curve} \rightarrow \text{Exponential recovery fit} \rightarrow \text{FRAP parameters}.$$

2 Input Intensities

For each sample, the input dataframe contains a time series of target and control fluorescence intensities. The target intensity is denoted by

$$I_{\text{target}}(t),$$

and the control intensity is denoted by

$$I_{\text{control}}(t).$$

The pre-bleach reference time is

$$t_{\text{pre}},$$

which corresponds in the implementation to

$$t_{\text{pre}} = \texttt{PREBLEACH_TIME}.$$

The pre-bleach frame is identified by the condition

$$t = t_{\text{pre}}.$$

If no such frame exists, the sample cannot be corrected and is rejected.

3 Photobleaching Correction

The control channel is used to estimate global photobleaching during image acquisition. First, the pre-bleach control intensity is extracted:

$$I_{\text{control,pre}} = I_{\text{control}}(t_{\text{pre}}).$$

The control bleaching factor is then computed at each time point:

$$B_{\text{control}}(t) = \frac{I_{\text{control}}(t)}{I_{\text{control,pre}}}.$$

The corrected target intensity is obtained by dividing the measured target intensity by the control bleaching factor:

$$I_{\text{target,corr}}(t) = \frac{I_{\text{target}}(t)}{B_{\text{control}}(t)}.$$

Equivalently,

$$I_{\text{target,corr}}(t) = I_{\text{target}}(t) \cdot \frac{I_{\text{control,pre}}}{I_{\text{control}}(t)}.$$

This correction compensates for acquisition-related intensity loss observed in the control region. If

$$I_{\text{control,pre}} = 0,$$

the correction is undefined and the sample is rejected.

4 Corrected Pre-Bleach Target Intensity

After photobleaching correction, the corrected pre-bleach target intensity is stored as

$$I_{\text{pre}} = I_{\text{target,corr}}(t_{\text{pre}}).$$

This value is later used for percentage normalization and for estimating the mobile fraction.

5 Per-Sample Percentage Normalization

Each corrected target curve can be normalized to its own corrected pre-bleach intensity:

$$I_{\text{target,corr,\%}}(t) = \frac{I_{\text{target,corr}}(t)}{I_{\text{pre}}} \cdot 100.$$

Therefore, for each valid sample,

$$I_{\text{target,corr,\%}}(t_{\text{pre}}) = 100.$$

If I_{pre} is zero or not finite, the percent-normalized curve is undefined.

6 Time Filtering

Only frames from the configured minimum plotting time onward are retained:

$$t \geq t_{\min},$$

where

$$t_{\min} = \texttt{MIN_PLOT_TIME}.$$

This step removes frames that should not be included in plotting or downstream recovery analysis.

7 Selection of the Fitted Signal

The fitted signal depends on the selected normalization mode.

7.1 Raw Corrected Mode

If

$$\texttt{NORMALIZATION_MODE} = \texttt{raw_corrected},$$

then the fitted signal is

$$y(t) = I_{\text{target,corr}}(t).$$

The pre-bleach intensity used for mobile fraction estimation is the group mean of the per-sample corrected pre-bleach intensities:

$$I_{\text{pre,fit}} = \frac{1}{N} \sum_{j=1}^{N} I_{\text{pre},j},$$

where N is the number of samples in the group.

7.2 Percent of Own Pre-Bleach Mode

If

$$\texttt{NORMALIZATION_MODE} = \texttt{percent_of_own_prebleach},$$

then the fitted signal is

$$y(t) = I_{\text{target,corr},\%}(t).$$

The pre-bleach intensity used for mobile fraction estimation is fixed to

$$I_{\text{pre,fit}} = 100.$$

This is because each sample is normalized such that its own corrected pre-bleach intensity equals 100%.

8 Group Mean Recovery Curve

For each experimental group, the selected signal $y(t)$ is aggregated across samples. At each time point, the group mean recovery curve is computed as

$$\bar{y}(t) = \frac{1}{N_t} \sum_{j=1}^{N_t} y_j(t),$$

where N_t is the number of samples contributing to that time point.

The group mean curve $\bar{y}(t)$ is the curve fitted by the exponential recovery model.

9 One-Phase Exponential Recovery Model

The group mean FRAP recovery curve is modeled using a one-phase exponential recovery function:

$$I(t) = I_\infty - Ae^{-kt}.$$

Here,

$$I_\infty$$

is the fitted recovery plateau,

$$A$$

is the fitted recovery amplitude, and

$$k$$

is the fitted recovery rate constant.

The fitting time axis is shifted so that the first fitted frame corresponds to zero:

$$t_{\text{start}} = \min(x_i),$$

$$t_i = x_i - t_{\text{start}},$$

where x_i are the original time values.

The fitted model is therefore

$$\bar{y}_i \approx I_\infty - Ae^{-kt_i}.$$

10 Initial Parameter Estimates

Before nonlinear curve fitting, the initial parameter vector is estimated from the group mean curve. The initial plateau estimate is the final observed group mean intensity:

$$I_\infty^{(0)} = \bar{y}_{\text{last}}.$$

The initial amplitude estimate is

$$A^{(0)} = \max\left(I_\infty^{(0)} - \bar{y}_0, 10^{-6}\right).$$

The initial recovery rate constant is set to the configured value:

$$k^{(0)} = k_{\text{init}}.$$

Thus, the initial parameter vector is

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} I_\infty^{(0)} \\ A^{(0)} \\ k^{(0)} \end{bmatrix}.$$

11 Parameter Bounds

The nonlinear fit constrains the parameters to physically meaningful ranges:

$$I_\infty \geq 0,$$

$$A \geq 0,$$

$$k \geq k_{\min}.$$

12 Fitted Initial Post-Bleach Intensity

At shifted time $t = 0$, the model gives

$$I(0) = I_\infty - A.$$

The fitted initial post-bleach intensity is therefore

$$I_0 = I_\infty - A.$$

This value represents the model-estimated fluorescence intensity immediately after bleaching.

13 Recovery Half-Time

The recovery half-time is the time required for the fitted recovery curve to recover half of its amplitude. For

$$I(t) = I_\infty - Ae^{-kt},$$

the half-time is

$$t_{1/2} = \frac{\ln 2}{k}.$$

A larger value of k corresponds to faster recovery and therefore to a smaller value of $t_{1/2}$. The half-time on the original, unshifted time axis is

$$t_{1/2}^{\text{abs}} = t_{\text{start}} + t_{1/2}.$$

14 Half-Recovery Intensity

The intensity at half recovery is halfway between I_0 and I_∞:

$$I_{1/2} = I_0 + \frac{A}{2}.$$

Since

$$A = I_\infty - I_0,$$

this can also be written as

$$I_{1/2} = \frac{I_0 + I_\infty}{2}.$$

15 Mobile Fraction

The total bleached amount is estimated as

$$I_{\text{pre,fit}} - I_0.$$

The recovered amount is the fitted recovery amplitude:

$$A = I_\infty - I_0.$$

The mobile fraction is therefore

$$M_f = \frac{A}{I_{\text{pre,fit}} - I_0}.$$

Equivalently,

$$M_f = \frac{I_\infty - I_0}{I_{\text{pre,fit}} - I_0}.$$

The mobile fraction is clipped to the interval

$$0 \le M_f \le 1.$$

If

$$I_{\text{pre,fit}} - I_0 \le 0,$$

or if the denominator is not finite, then the mobile and immobile fractions are undefined.

16 Immobile Fraction

The immobile fraction is defined as the unrecovered fraction:

$$I_f = 1 - M_f.$$

A high mobile fraction indicates that most of the bleached signal recovered, whereas a high immobile fraction indicates incomplete recovery.

17 Smooth Fitted Recovery Curve

For visualization, the fitted model is evaluated over a dense set of shifted time points:

$$t_{\text{smooth}} \in [\min(t_i), \max(t_i)].$$

The fitted intensity values are

$$I_{\text{fit}}(t_{\text{smooth}}) = I_\infty - Ae^{-kt_{\text{smooth}}}.$$

The smooth time axis is then shifted back to the original time coordinates:

$$x_{\text{smooth}} = t_{\text{smooth}} + t_{\text{start}}.$$

The output fitted curve is therefore represented as

$$\left(x_{\text{smooth}}, I_{\text{fit}}(t_{\text{smooth}})\right).$$

18 Final Output Per Group

For each group, the pipeline returns the following quantities:

Quantity	Meaning
group	Group identifier
normalization mode	Selected normalization mode
$I_{\text{pre,fit}}$	Pre-bleach reference intensity used for mobile fraction estimation
I_0	Fitted initial post-bleach intensity
I_∞	Fitted recovery plateau
A	Fitted recovery amplitude
k	Fitted recovery rate constant
$t_{1/2}$	Recovery half-time on the shifted fitting time axis
$t_{1/2}^{\text{abs}}$	Recovery half-time on the original time axis
$I_{1/2}$	Intensity at half recovery
M_f	Mobile fraction
I_f	Immobile fraction
n_{files}	Number of unique source files contributing to the group

Table 1: Summary of output quantities generated for each fitted group.

19 Conceptual Interpretation

The analysis estimates how fluorescence in a bleached target region recovers over time after correcting for global photobleaching using a control region.

The control signal is used to compensate for acquisition-related intensity loss:

$$\text{Control channel} \rightarrow \text{photobleaching correction.}$$

The corrected target signal is then optionally normalized relative to its own pre-bleach intensity:

$$\text{Target channel} \rightarrow \text{corrected and normalized recovery curve.}$$

The corrected recovery curves are averaged within each group:

$$\text{Corrected recovery curves} \rightarrow \text{group mean recovery curve.}$$

The group mean curve is fitted using a one-phase exponential recovery model:

$$\text{Group mean recovery curve} \rightarrow I(t) = I_{\infty} - Ae^{-kt}.$$

Finally, the fitted model is used to estimate recovery kinetics and recovery extent:

$$\text{Exponential fit} \rightarrow \left\{ I_0, I_{\infty}, A, k, t_{1/2}, M_f, I_f \right\}.$$